



Externally set task 2016

Student Number: In figures

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In words

HUMAN BIOLOGY GENERAL COURSE

Students can use the following items for this task

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

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Total time for the task: 50 minutes
Total marks: 50 marks
Weighting: 15% of the school mark

Question 1**(23 marks)**

A study was undertaken into the factors affecting the reaction times of drivers. Three young drivers (Ella, Hiro and Joseph) who had recently obtained their drivers licences, participated in the study. Results recorded in the study included the amount of time it took for the participants to react to a hazard (reaction time) under various conditions that could cause driver impairment. The results of the study are shown below.

Driver reaction time under various conditions

Travelling at an average driving speed of 60 km/h	Reaction time (s)			
	Ella	Hiro	Joseph	Mean
Baseline (under normal conditions)	0.45	0.49	0.54	0.49
Reading a text message	0.79	0.83	1.12	
Writing a text message	0.91	0.86	1.04	
At 0.08 blood alcohol concentration	0.76	0.79	0.94	

The following questions refer to the data from the study outlined above.

- (a) Formulate a hypothesis for the study. (2 marks)

- (b) Calculate the mean reaction times for reading a text message, writing a text message and at 0.08 blood alcohol concentration. Insert the answer into the blank cells in the table above. (3 marks)

- (c) The data on reaction time were recorded in seconds to two decimal places. Explain why this was used as the unit for measurement rather than minutes. (2 marks)

- (d) Suggest **three** possible variables that would have been controlled for the study to be a fair test. (3 marks)

One: _____

Two: _____

Three: _____

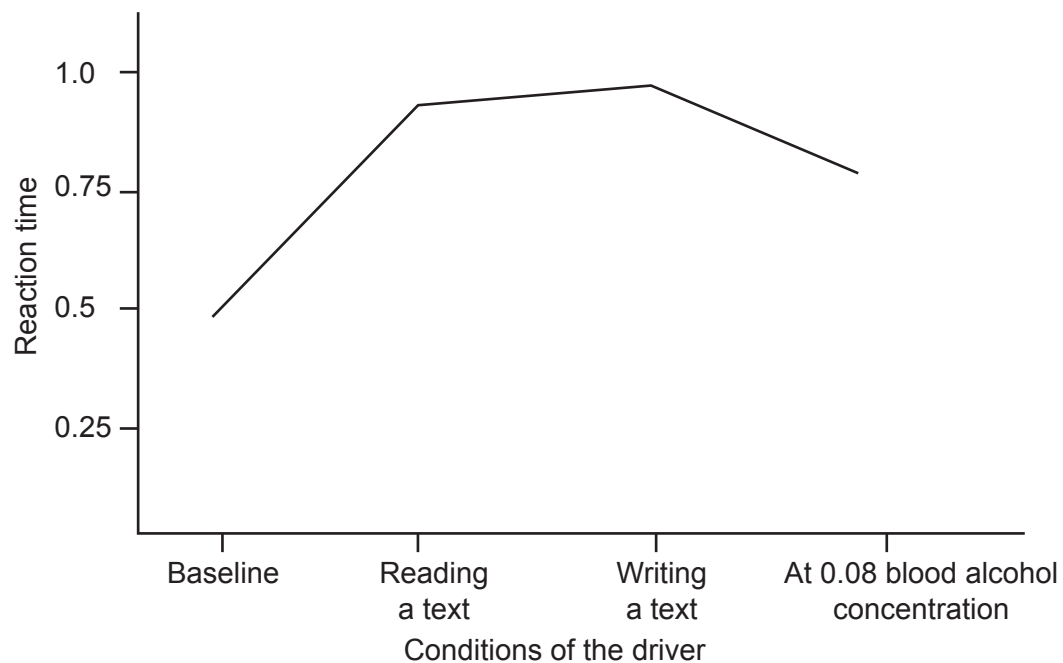
- (e) Why was the baseline measurement of reaction time taken? (1 mark)

- (f) Suggest **one** way in which the study could be modified to increase its

- (i) reliability. (1 mark)

- (ii) validity. (1 mark)

- (g) One of the students attempted to graph the mean results. The graph is shown below.



There are three major errors in the graph above. Identify the **three** errors. (3 marks)

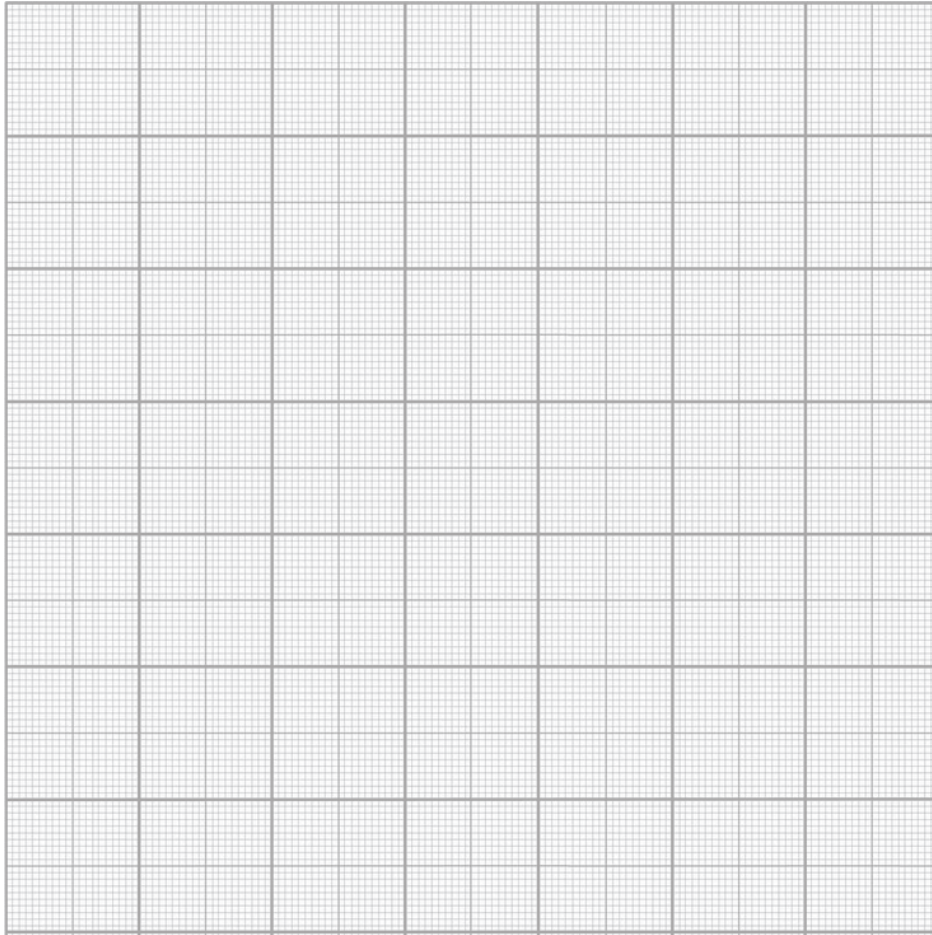
One: _____

Two: _____

Three: _____

- (h) Graph the mean data from the table correctly.

(5 marks)

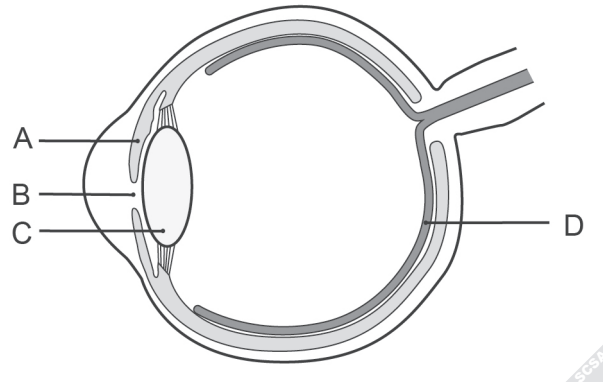


- (i) With reference to the data obtained from the study, describe a reasonable conclusion that can be drawn. (2 marks)

Question 2**(14 marks)**

Numerous hazards occur during every road journey. The body must detect, interpret and react to these hazards to ensure that no accidents occur. This requires the integration of many systems. The following questions refer to these systems.

One of the first steps in reacting to a hazard is its visual detection by the eye. Parts (a) and (b) of the question refer to the diagram of the eye shown below.



- (a) For the structures listed below, outline the function of each as it relates to the detection of visual stimuli. (3 marks)

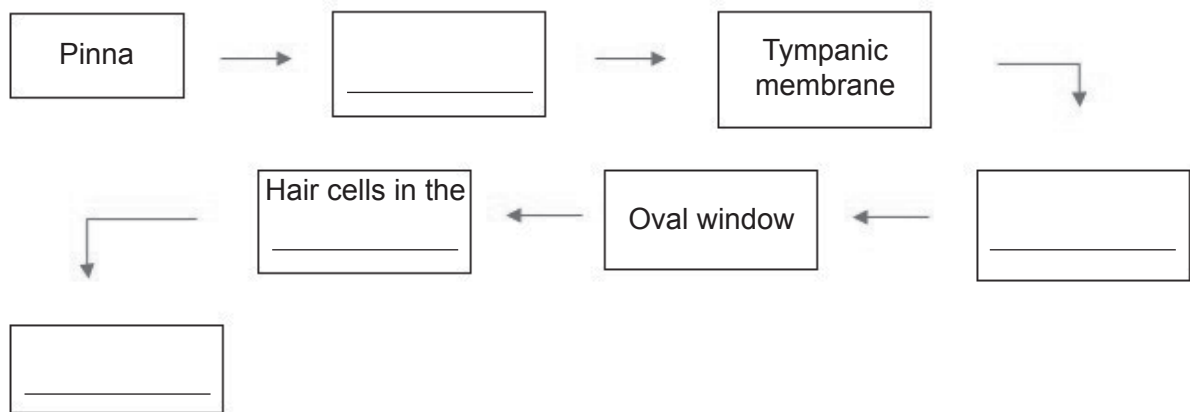
A: _____

B: _____

C: _____

- (b) Describe how light hitting structure D is transmitted to the brain. (2 marks)

Another structure in the body that can detect road hazards is the ear. The structures of the ear through which sound must pass through to reach the brain are represented by the diagram below.



(c) (i) Complete the diagram above by identifying the missing structures. (4 marks)

(ii) Indicate by circling the structure on the diagram above the point that sound energy is converted into mechanical energy during its passage through the ear. (1 mark)

(iii) How is sound energy converted to mechanical energy by the structure identified in part (c)(ii)? (1 mark)

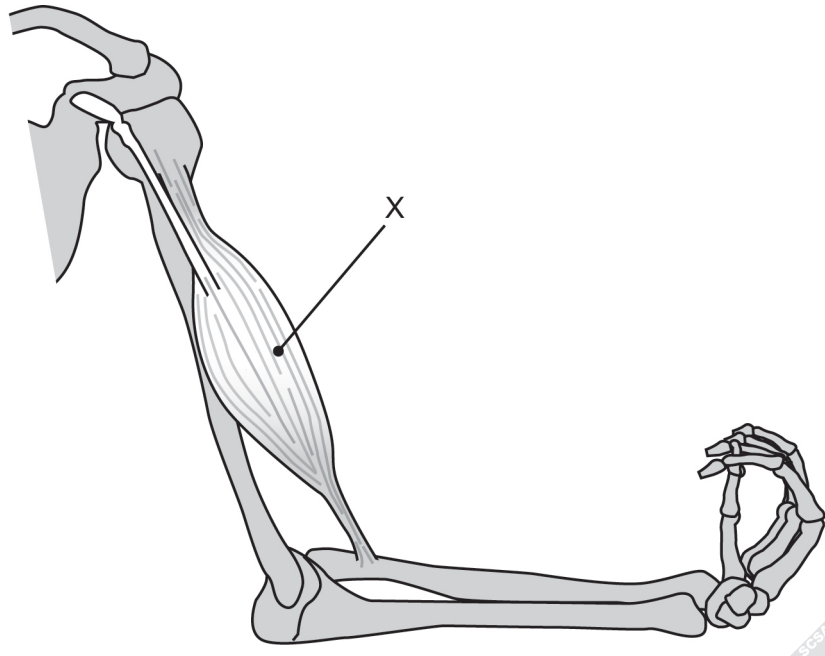
(d) Once the stimuli detected by the eyes and ears reach the brain, they must be processed. Explain how the brain brings about a response in the muscles of the body. (3 marks)

Question 3

(13 marks)

One reaction to a road hazard is to swerve the car by moving the steering wheel. This movement involves the muscles and the bones of the arm working in a coordinated action.

Parts (a) and (b) of the question refer to the diagram of the arm shown below.



- (a) (i) Several joints are shown in the diagram. Circle **two** different types of joints on the diagram. (2 marks)

- (ii) Name each joint you have circled, state its type and describe the kind of movement the joint allows. (4 marks)

One: _____

Two: _____

- (b) Describe how the contraction of muscle shown as X in the upper arm creates movement in the lower arm. (4 marks)

- (c) Explain how muscles work as antagonistic pairs and name **one** example of an antagonistic pair of muscles in the human body. (3 marks)

A large, light grey, stylized leaf graphic that occupies the left and bottom-left portion of the page. It has a central vein and several smaller veins branching out, resembling a fan or a large leaf.

HUMAN BIOLOGY

GENERAL COURSE

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Question 1

(23 marks)

- (a) Formulate a hypothesis for the study.

(2 marks)

Description	Marks
Testable statement	1
Incorporates the independent and dependent variables.	1
Total	2
Answer could include, but is not limited to:	
E.g. Texting while driving significantly increases reaction time.	

- (b) Calculate the mean reaction times for reading a text message, writing a text message and at 0.08 blood alcohol concentration. Insert the answer into the blank cells in the table above. (3 marks)

Description	Marks
Reading – 0.91	1
Writing - 0.94	1
Blood alcohol – 0.83	1
Total	3

- (c) The data on reaction time were recorded in seconds to two decimal places. Explain why this was used as the unit for measurement rather than minutes. (2 marks)

Description	Marks
Reaction time is very fast/doesn't take a long time	1
If recorded in minutes the numbers would be very small/hard to work with	1
Total	2

- (d) Suggest
- three**
- possible variables that would have been controlled for the study to be a fair test. (3 marks)

Description	Marks
Any three of:	
• same road conditions • same weather conditions • same hazard • same vehicle • similar driving experience/ability • similar medical history/health	1–3
Total	3

- (e) Why was the baseline measurement of reaction time taken?

(1 mark)

Description	Marks
To use as a control/so that the experimental data can be compared	1
Total	1

(f) Suggest **one** way in which the study could be modified to increase its

(i) reliability. (1 mark)

Description	Marks
Repeat experiment again/replicate/increase sample size/ specific answer referring to the experiment	1
Total	1

(ii) validity. (1 mark)

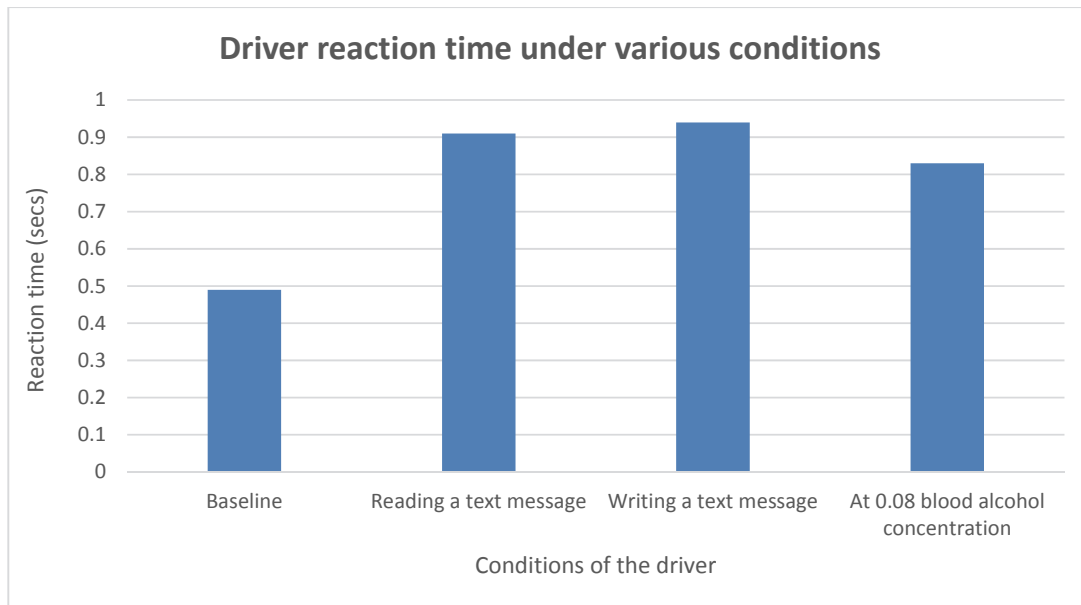
Description	Marks
Establish more controlled variables/improve experimental design/ specific answer referring to the experiment	1
Total	1

(g) One of the students attempted to graph the mean results. The graph is shown below. There are three major errors in the graph above. Identify the **three** errors. (3 marks)

Description	Marks
No title	1
Line graph instead of a column graph	1
No units for reaction time axis	1
Total	3

- (h) Graph the mean data from the table correctly.

(5 marks)



Description	Marks
Correctly plots points as a column graph (with spaces between columns)	1
Labelling of axes with correct name and unit	1
Uses a suitable scale	1
Title appropriate with both variables included	1
Uses a ruler	1
Total	5
Note: no points deducted if the mean scores were calculated wrong initially	

- (i) With reference to the data obtained from the study, describe a reasonable conclusion that can be drawn. (2 marks)

Description	Marks
Reasonable concluding statement	1
Incorporation of data into conclusion	1
Total	2
Answer could include, but is not limited to: Texting while driving is more dangerous than drink driving Reaction time while texting is 0.08-0.11 of a second slower than when driving drunk (or any other reasonable data from the table)	

Question 2

(14 marks)

- (a) For the structures listed below, outline the function of each as it relates to the detection of visual stimuli. (3 marks)

Description	Marks
A - Changes the size of the pupil	1
B - Allows light into the eye	1
C - Focuses light onto the retina	1
Total	3

- (b) Describe how light hitting structure D is transmitted to the brain. (2 marks)

Description	Marks
D converts the light into an electrical signal	1
Message transmitted along the optic nerve to the brain	1
Total	2

- (c) (i) Complete the diagram above by identifying the missing structures. (4 marks)

Description	Marks
Auditory Canal	1
Ossicles (or malleus, incus, and stapes)	1
Cochlea	1
Auditory nerve	1
Total	4

- (ii) Indicate by circling the structure on the diagram above the point that sound energy is converted into mechanical energy during its passage through the ear. (1 mark)

Description	Marks
Tympanic membrane	1
Total	1

- (iii) How is sound energy converted to mechanical energy by the structure identified in part (c)(ii)? (1 mark)

Description	Marks
It causes the membrane to beat/vibrate back and forth	1
Total	1

- (d) Once the stimuli detected by the eyes and ears reach the brain, they must be processed. Explain how the brain brings about a response in the muscles of the body. (3 marks)

Description	Marks
Any three of:	
- information processed by the cerebrum - message passes through the cerebellum to ensure smooth muscle movement - cerebrum initiates nervous impulses to voluntary muscles - muscles of the arms/legs are stimulated to contract	1–3
Total	3

Question 3

(13 marks)

- (a) (i) Several joints are shown in the diagram. Circle **two** different types of joints on the diagram. (2 marks)

Description	Marks
Any two of:	1–2
• wrist (pivot or condyloid joint)	
• elbow/phalanges (hinge joints)	
• knuckles (condyloid joints)	
• shoulder (ball and socket joint)	
• thumb (saddle joint)	
Total	2
Note: Must clearly circle the joint and not the tendon attachment. Cannot have multiple joints in one circle. Must circle two different types of joints for one mark each.	

- (ii) Name each joint you have circled, state its type and describe the kind of movement the joint allows. (4 marks)

Description	Marks
One mark for the name and one mark for the type of movement	1–4
Any two of:	
• Condyloid joint, allows flexion and extension and sideways (two planes) movement	
• Ball and socket joint, allows movement front to back and side to side and rotational (three planes)/movement in all directions	
• Hinge, allows flexion and extensions (one plane) movement	
• Saddle, allows side to side or back and forth movement (two planes)	4
• Pivot joint, allows rotational movement	
Total	4
Note: Must match the two joints circled on the diagram	

- (b) Describe how the contraction of muscle shown as X in the upper arm creates movement in the lower arm. (4 marks)

Description	Marks
Muscle X shortens	1
Pulls on the bones of the lower arm	1
Creates movement at the elbow joint	1
Flexion (lower arm moves towards bicep/angle reduced) of arm results	1
Total	4

- (c) Explain how muscles work as antagonistic pairs and name **one** example of an antagonistic pair of muscles in the human body. (3 marks)

Description	Marks
One moves the bone in one direction and the other moves it back the other way/move bones in opposite directions	1
One muscle must contract while the other relaxes	1
Example	1
Total	3
Answer could include, but is not limited to:	
Possible examples: biceps and triceps/back (deltoid muscle or latissimus dorsi) and chest (pectoralis major)/abdominals and lower back/gluteus maximus and lower abdominal/quadriceps and hamstrings/shin (tibialis anterior) and calf (soleus/gastrocnemius)	

